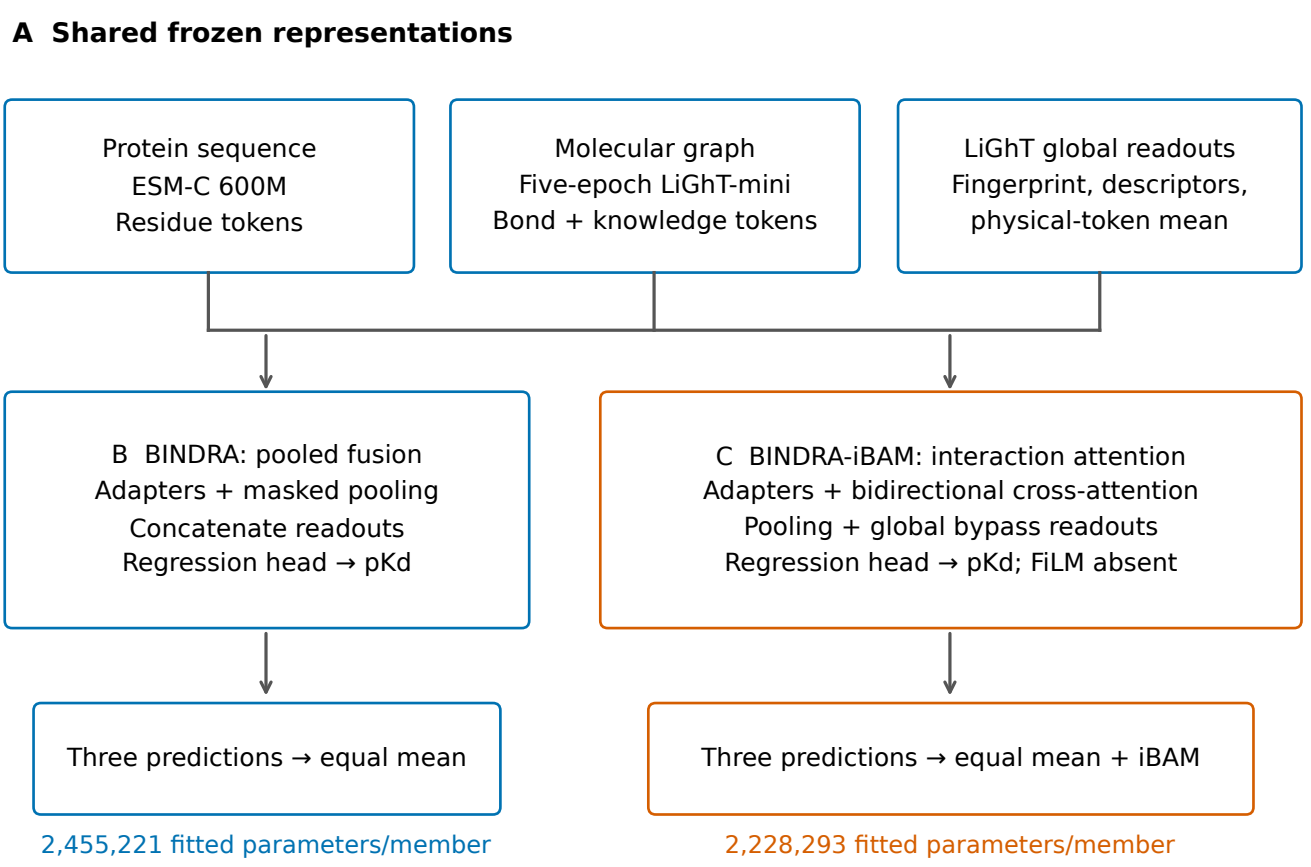
A Shared frozen representations
Protein sequence
ESM-C 600M
Residue tokens
Molecular graph
Five-epoch LiGhT-mini
Bond + knowledge tokens
LiGhT global readouts
Fingerprint, descriptors,
physical-token mean
B BINDRA: pooled fusion
Adapters + masked pooling
Concatenate readouts
Regression head → pKd
C BINDRA-iBAM: interaction attention
Adapters + bidirectional cross-attention
Pooling + global bypass readouts
Regression head → pKd; FiLM absent
Three predictions → equal mean
Three predictions → equal mean + iBAM
2,455,221 fitted parameters/member
2,228,293 fitted parameters/member